

Arya Shrestha

551-229-7377 | arya.shrestha1@marist.edu | [linkedin.com/in/aryashrestha05](https://www.linkedin.com/in/aryashrestha05) | github.com/AryaShrestha05

EDUCATION

Marist University

Poughkeepsie, NY

Bachelor of Science in Computer Science **CGPA: 4.0**

Expected May 2028

- **Relevant Courseworks:** Data Structures & Algorithms, Object-Oriented Programming, Software Development I & II, Database Management (SQL), Discrete Mathematics

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, SQL (PostgreSQL), R

Developer Tools: Git & GitHub, Docker, Kubernetes

Additional Skills: Agile & Scrum, SDLC, Unit, Integration & End-to-end Testing, NoSQL (MongoDB)

EXPERIENCE

Software Engineer Intern

June 2025 – September 2025

Koding 4 Kids

St Peter, MN

- Built and shipped a gamified, AI-powered learning platform serving **50+** active users across schools and libraries, achieving **95%+** retention in pilot deployments.
- Designed and implemented modular **REST APIs** for progress tracking and AI tutoring, directly contributing to **\$9,000** in secured funding for statewide expansion.
- Developed an **automated** SMS and email notification pipeline, **improving engagement** and retention in under-served communities.

LMS Quality Assurance Developer

April 2025 – Present

Marist University

Poughkeepsie, NY

- Designed and developed a **150+ page website** for the Marist Digital Education department's initiatives.
- Tracked, documented, and escalated system defects using **Jira** and **TeamDynamix**, collaborating with engineers and stakeholders to resolve platform issues.
- Audited and validated accessibility compliance (WCAG 2.1 AA) across **3,000+** LMS courses using automated tools and manual verification, reducing institutional compliance risk.

PROJECTS

Legacy Louisiana | Node JS, Express JS, PostgreSQL, React, Tailwind

July 2025 – December 2025

- Developed a **full-stack web application to automate trust and will document creation**, allowing attorneys and clients in the state of Louisiana to securely input legal data via a PostgreSQL-backed form interface.
- Projected to reduce manual workload by over **95%**, **decreasing attorney drafting time from 6 hours to under 30 minutes** by generating **structured PDF summaries with embedded charts and visual data**.
- Engineering a scalable system architecture that maintains full data ownership, supports **projected future user growth from 60 to 1000+**.

Posturize | React JS, Flask, OpenCV, MediaPipe, D3.js

November 2025, **HackPrinceton**

- Engineered a **full-stack, real-time** posture analysis system with accurate **vision-based pose detection** and **REST APIs** for rapid feedback in under **36 hours**.
- Integrated **OpenCV and MediaPipe** to extract body landmarks and compute posture metrics in real time.
- Created real-time D3.JS visualizations to deliver instant feedback that improved overall user engagement

Polynomial Regression Model from Scratch | Python, Matplotlib, Numpy

Dec. 2024 – Jan. 2025

- Built and optimized a polynomial regression algorithm to predict housing prices using an open-source Kaggle database with **5000+** data points.
- Implemented gradient descent to minimize the cost function, improving prediction accuracy by **15 Percent**.
- **Feature-engineered** variables from existing data, enhancing model interpretability and performance.

CERTIFICATION & INVOLVEMENTS

Deeplearning.ai Machine Learning Classification, **ColorStack Fellow**, **AI4ALL Ignite Fellow**, **Notion Campus Leader**, Marist Computer Science Society, Marist Analytics Club, **HackPrinceton**